Emilio Fayad Villarreal

608-960-2380 | emiliofav@gmail.com | linkedin.com/in/emilio-fayad | github.com/efayadv

Education

University of Wisconsin - Madison Madison, WI

Bachelor of Science in Computer Science *Sep. 2022 – May 2026*

- **Relevant Coursework:** Algorithms, Software Engineering, Database Management, Artificial Intelligence, Operating Systems, Data Structures, Compilers, Linear Algebra for Machine Learning, Applied Statistics

Experience

AI Software Engineer Intern June 2025 – Aug 2025

Eciton Global *Monterrey, Mexico*

- Built a Python pipeline using YOLO, OCR, and OpenCV to convert retail shelf images into structured data.
- Developed preprocessing and ETL components for 12 client formats, achieving ∼85% extraction accuracy and reducing manual analysis and operating costs by ∼25%.
- Integrated JSON outputs with PostgreSQL to store extracted product data and image metadata for downstream analysis.
- Enabled natural-language querying of database using retrieval-augmented generation (RAG) workflows with LlamaIndex, OpenAI APIs, embeddings, and vector search.

Software Engineer - Research Assistant Oct 2024 – March 2025

University of Wisconsin - Madison *Madison, WI*

- Built a real-time translation tool to enable bilingual classroom communication with a Python/Flask application using PyAudio, Google Speech-to-Text, Google Translate, and Whisper AI.
- Achieved ∼1-second response time supported by implementing API validation, latency monitoring, and error recovery, integrated with Google Cloud Storage for audio persistence.
- Enhanced model accuracy from 80% to 94% while supporting multilingual and noisy classroom conditions using collected audio transcriptions from K-12 classrooms to fine-tune Whisper AI.

Data Analyst Intern June 2023 – August 2023

Fair Play Retail *Monterrey, Mexico*

- Helped increase gross profit margin from 12% to 19% by analyzing trending sales and price adjustments with Excel
- Negotiated bulk pricing with manufacturers including P&G and Kimberly-Clark, presenting Power BI dashboards to support purchasing proposals and secure lower prices.

Projects

Glance - Mobile App | *TypeScript, PostgreSQL, React Native* June 2026 – Present

- Built using React Native, TypeScript, Supabase, and PostgreSQL to enable venue discovery and crowdsourced wait-time reporting.
- Designed PostgreSQL schemas in Supabase for venue reports and anonymous posts, using Row-Level Security (RLS) policies to restrict data access and protect user location data.
- Reviewed, debugged, and refactored AI-generated code from Cursor and Codex to accelerate development while preserving maintainability.

Villart - Web App | *JavaScript, PostgreSQL, Express, Vue.js* Jan 2025 – Feb 2025

- Built a registration web application serving 100+ monthly returning users using Vue.js, TypeScript, and Express.
- Designed PostgreSQL tables to manage class registrations, track payments, and enforce class capacity limits.
- Deployed the frontend to Vercel and the Express backend and REST APIs to Render.

Distributed Message Broker | *Java, ZooKeeper, TCP/IP, Maven* Aug 2026

- Built a Kafka-inspired distributing messaging system with custom binary TCP protocols using Java.
- Designed Zookeeper-based broker coordination and controller election using ephemeral nodes and watchers to support cluster and partition membership.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C, C++
Product Development: React Native, Node.js, Flask, REST APIs, mobile development
Backend: PostgreSQL, Supabase, Google Cloud Storage, authentication, relational data modeling, API integrations
Tools: Git, Maven, Travis CI, unit testing, integration testing, regression testing, debugging, structured logging